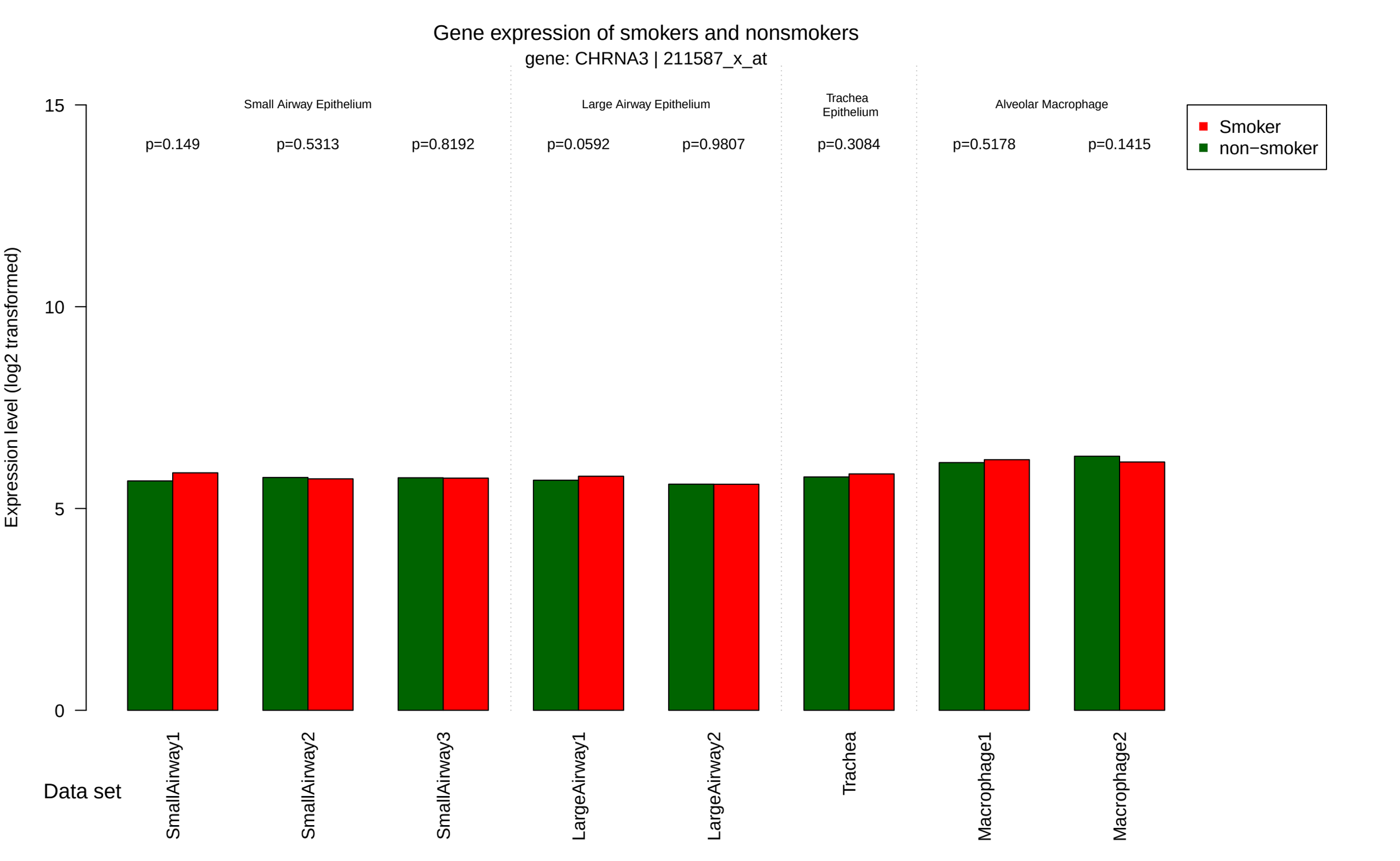
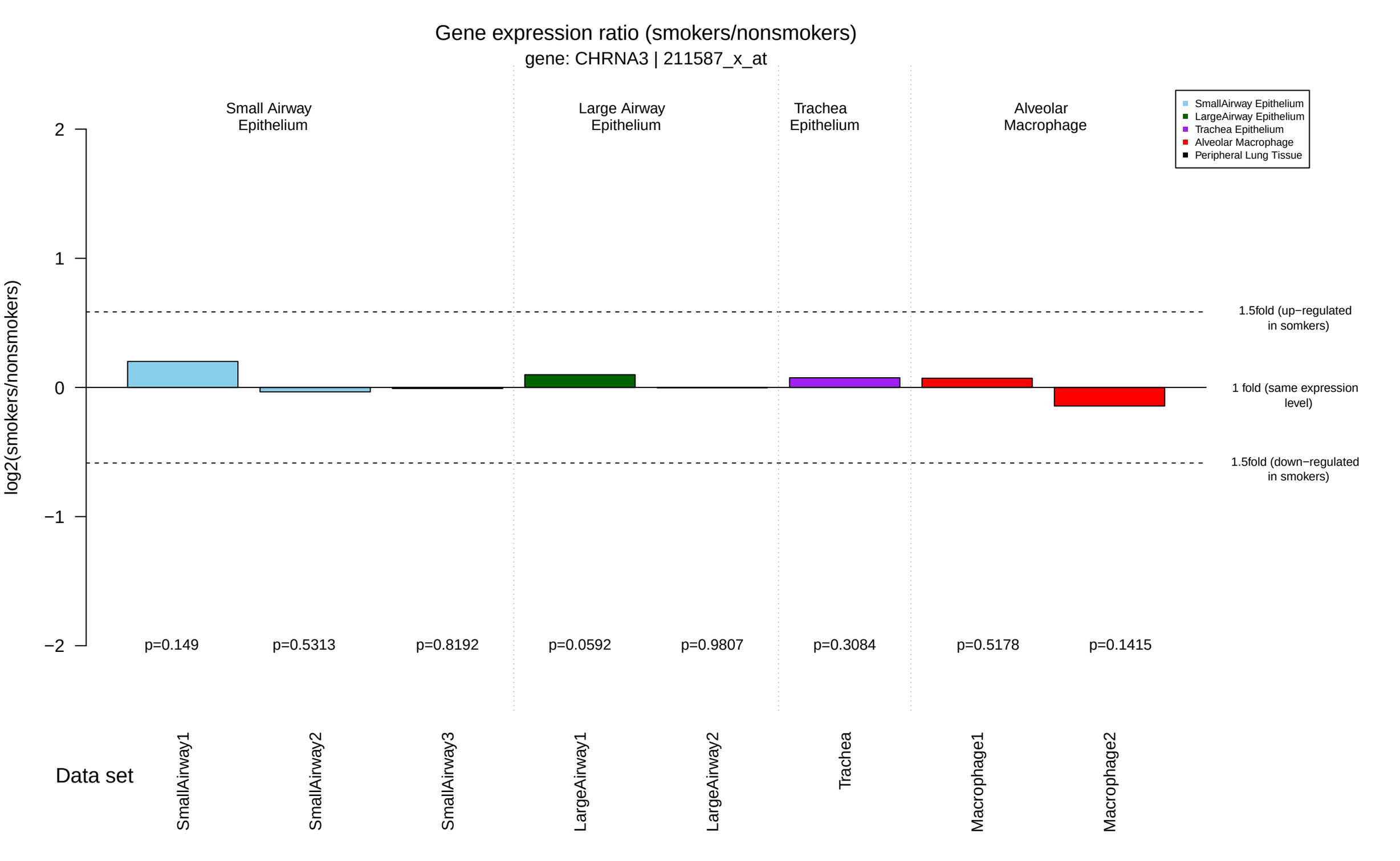
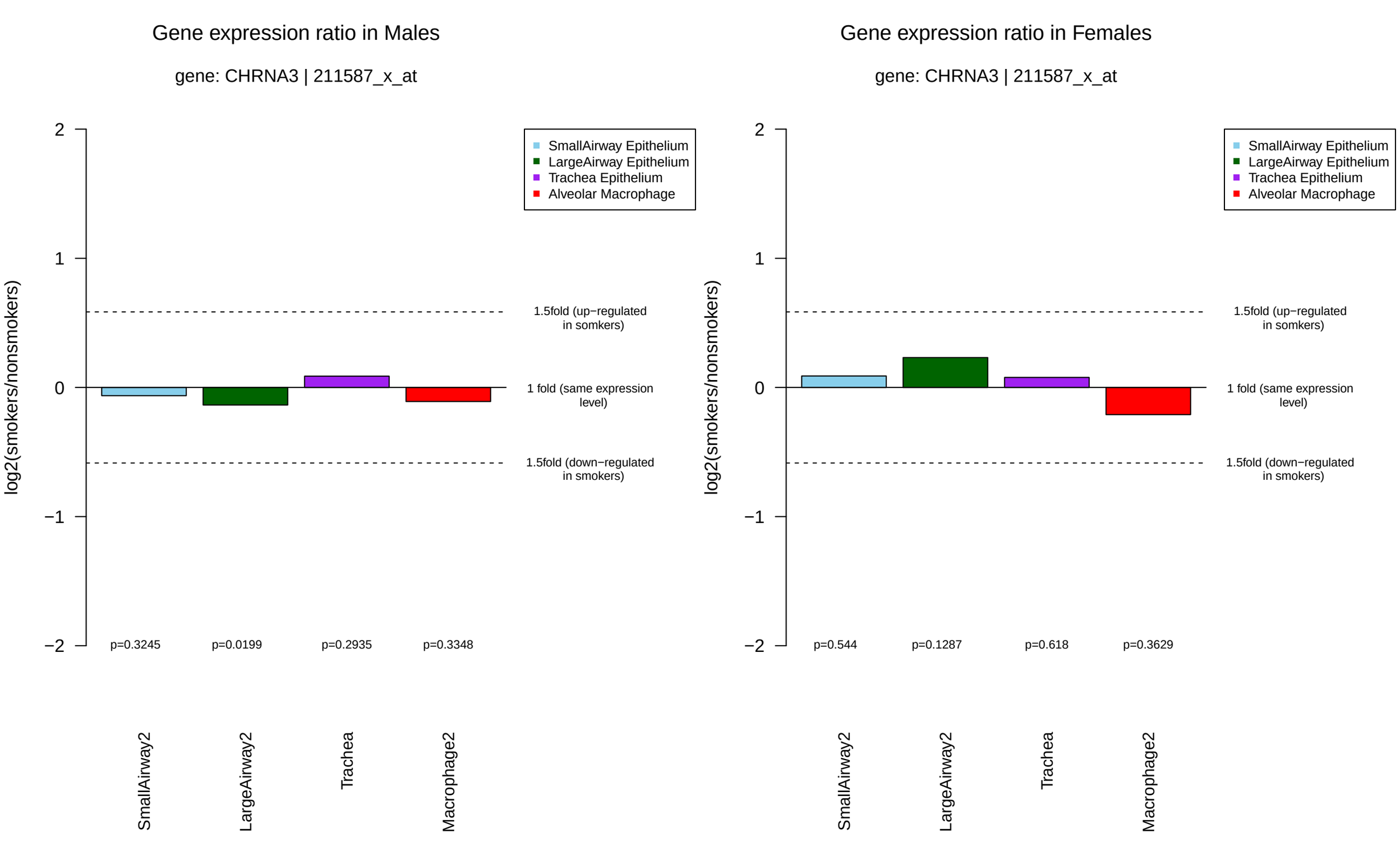
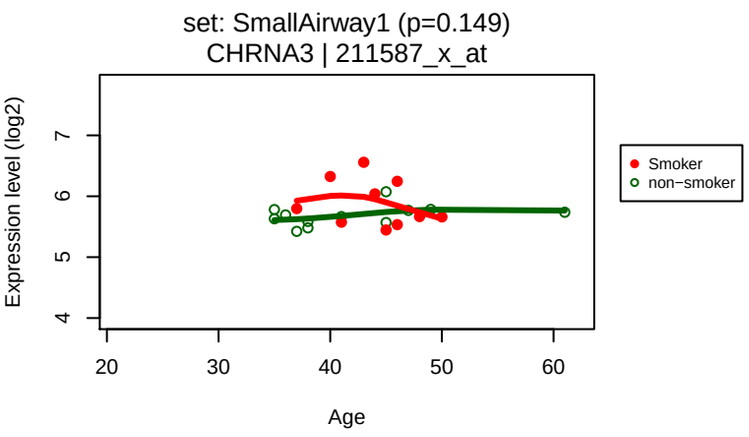
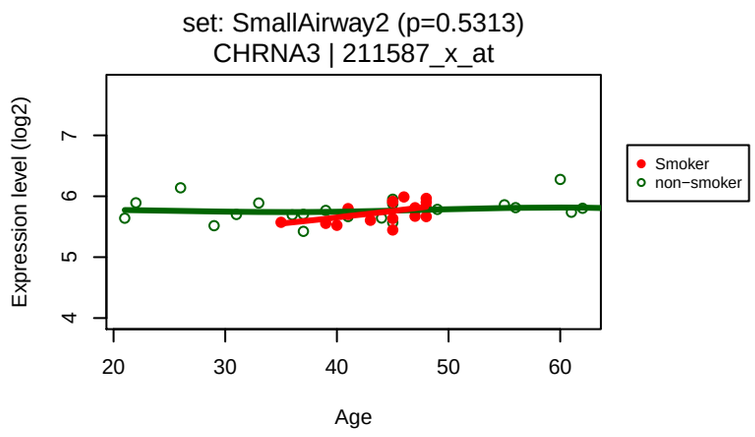
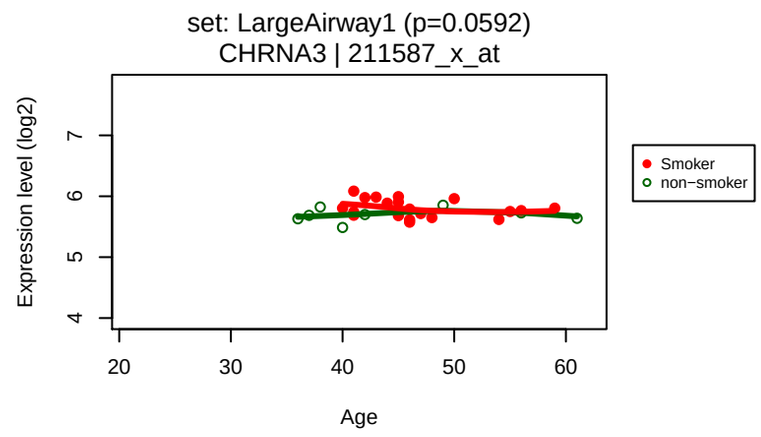

Gene expression of smokers and nonsmokers
gene: CHRNA3 | 211587_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.149
p=0.5313
p=0.8192
p=0.0592
p=0.9807
p=0.3084
p=0.5178
p=0.1415
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

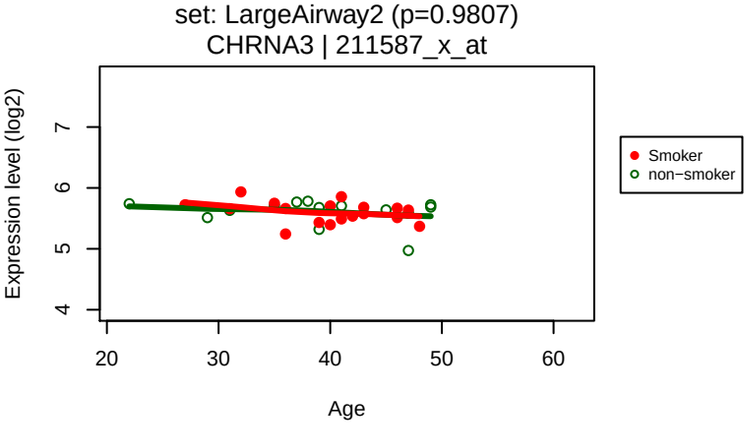

Gene expression ratio (smokers/nonsmokers)
gene: CHRNA3 | 211587_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.149
p=0.5313
p=0.8192
p=0.0592
p=0.9807
p=0.3084
p=0.5178
p=0.1415
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

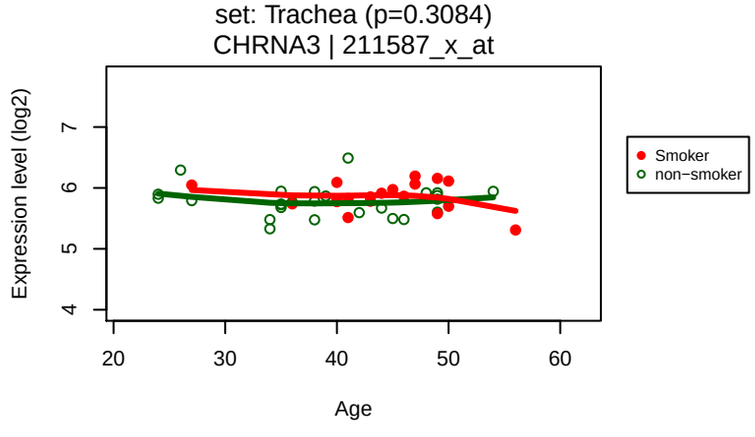

Gene expression ratio in Males
gene: CHRNA3 | 211587_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3245
p=0.0199
p=0.2935
p=0.3348
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CHRNA3 | 211587_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.544
p=0.1287
p=0.618
p=0.3629
SmallAirway2
LargeAirway2
Trachea
Macrophage2

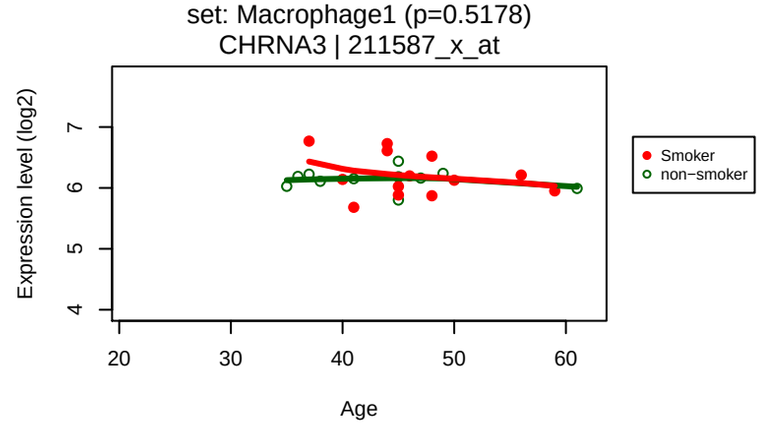

set: SmallAirway1 (p=0.149)
CHRNA3 | 211587_x_at
Expression level (log2)
Age
Smoker
non−smoker

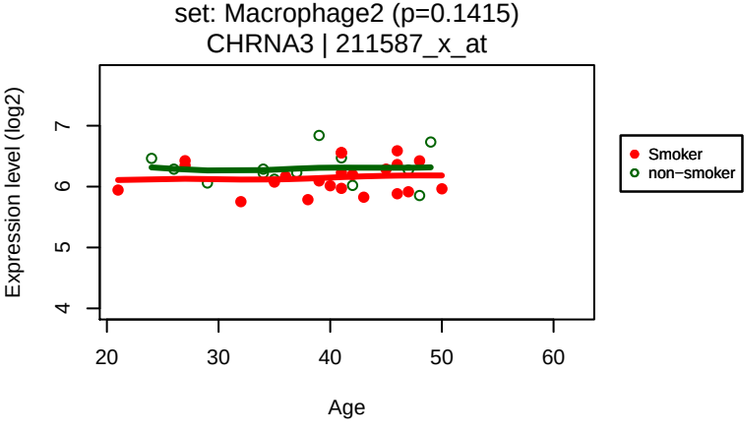

set: SmallAirway2 (p=0.5313)
CHRNA3 | 211587_x_at
Expression level (log2)
Age
Smoker
non−smoker

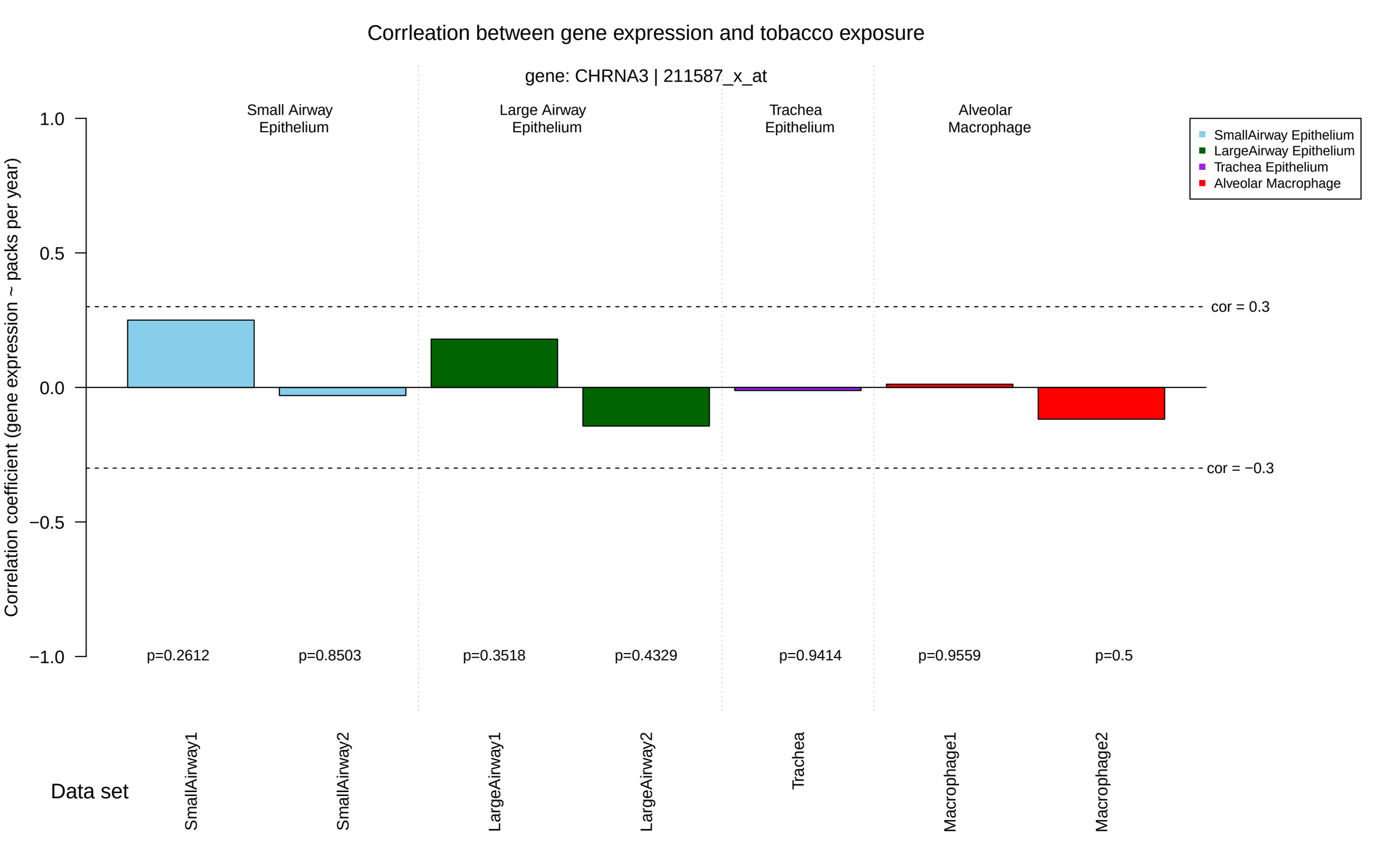

set: LargeAirway1 (p=0.0592)
CHRNA3 | 211587_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9807)
CHRNA3 | 211587_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3084)
CHRNA3 | 211587_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5178)
CHRNA3 | 211587_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1415)
CHRNA3 | 211587_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CHRNA3 | 211587_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2612
p=0.8503
p=0.3518
p=0.4329
p=0.9414
p=0.9559
p=0.5
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2